

2018 HSC Mathematics Marking Guidelines

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	A
4	D
5	D
6	C
7	C
8	D
9	B
10	D

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Attempts to use $3 - \sqrt{2}$, or equivalent merit	1

Sample answer:

$$\frac{3}{3 + \sqrt{2}} \times \frac{3 - \sqrt{2}}{3 - \sqrt{2}} = \frac{9 - 3\sqrt{2}}{9 - 2}$$

$$= \frac{9 - 3\sqrt{2}}{7}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Obtains $-3x > 9$, or equivalent merit	1

Sample answer:

$$1 - 3x > 10$$

$$\frac{-3x}{-3} > \frac{9}{-3}$$

$$x < -3$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Attempts to factorise numerator, or equivalent merit	1

Sample answer:

$$\frac{8x^3 - 27y^3}{2x - 3y} = \frac{(2x)^3 - (3y)^3}{2x - 3y}$$

$$= \frac{(2x - 3y)(4x^2 + 6xy + 9y^2)}{2x - 3y}$$

$$= 4x^2 + 6xy + 9y^2$$

Question 11 (d) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$a + 2d = 8 \quad \textcircled{1}$$

$$a + 19d = 59 \quad \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}: \frac{17d}{17} = \frac{51}{17}$$

$$d = 3$$

Question 11 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds first term, or equivalent merit	1

Sample answer:

$$d = 3, \text{ so } a + 2(3) = 8$$

$$a + 6 = 8$$

$$a = 2$$

$$a + 49d = 2 + 49 \times 3$$

$$= 2 + 147$$

$$= 149$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	2
• Provides correct primitive, or equivalent merit	1

Sample answer:

$$\int_0^3 e^{5x} dx = \left[\frac{1}{5} e^{5x} \right]_0^3$$

$$= \frac{1}{5} (e^{15} - e^0)$$

$$= \frac{1}{5} (e^{15} - 1)$$

Question 11 (f)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use product rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{d}{dx}(x^2 \tan x) &= x^2 \times \sec^2 x + \tan x \times (2x) \\ &= x(x \sec^2 x + 2 \tan x)\end{aligned}$$

Question 11 (g)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use quotient rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{d}{dx}\left(\frac{e^x}{x+1}\right) &= \frac{(x+1) \times e^x - e^x \times 1}{(x+1)^2} \\ &= \frac{xe^x + e^x - e^x}{(x+1)^2} \\ &= \frac{xe^x}{(x+1)^2}\end{aligned}$$

Question 12 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}\angle ABC &= 50^\circ + 60^\circ \\ &= 110^\circ\end{aligned}$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly substitutes into cosine rule, or equivalent merit	1

Sample answer:

$$d^2 = 320^2 + 190^2 - 2(320)(190)\cos 110^\circ$$

$$d = \sqrt{320^2 + 190^2 - 2(320)(190)\cos 110^\circ}$$

$$d = 424.3697084$$

$$d = 420 \text{ km}$$

Question 12 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains slope at $x = \frac{\pi}{6}$, or equivalent merit	2
• Obtains correct derivative, or equivalent merit	1

Sample answer:

$$y = \cos 2x$$

$$y' = -2\sin 2x$$

$$M_T = -2\sin\left(\frac{2\pi}{6}\right)$$

$$M_T = -2\sin\left(\frac{\pi}{3}\right)$$

$$= -\sqrt{3} \quad \text{at } x = \frac{\pi}{6}$$

$$\text{when } x = \frac{\pi}{6}, \quad y = \cos\left(\frac{2\pi}{6}\right)$$

$$= \frac{1}{2}$$

Equation of tangent

$$y - \frac{1}{2} = -\sqrt{3}\left(x - \frac{\pi}{6}\right)$$

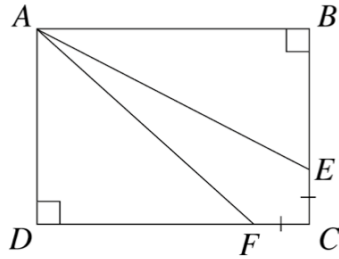
$$y - \frac{1}{2} = -\sqrt{3}x + \frac{\sqrt{3}\pi}{6}$$

$$y = -\sqrt{3}x + \frac{\sqrt{3}\pi}{6} + \frac{1}{2}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct proof	2
• Observes that $AD = AB$, with a reason, or equivalent merit	1

Sample answer:



In $\triangle ADF$ and $\triangle ABE$

$$AD = AB \quad (\text{equal sides of square } ABCD)$$

$$\angle ADF = \angle ABE = 90^\circ \quad (\text{equal angles of square } ABCD)$$

$$DF = DC - FC \quad (\text{equal sides of square } ABCD, FC = EC \text{ given})$$

$$= BC - EC$$

$$= BE$$

$$\therefore \triangle ADF \equiv \triangle ABE \quad (\text{SAS})$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds area of $\triangle ADF$, or equivalent merit	1

Sample answer:

$$\begin{aligned} \text{Area square} &= 14^2 \\ &= 196 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area triangle } ABE &= \frac{1}{2} \times 14 \times 10 \\ &= 70 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area } AECF &= 196 - (2 \times 70) \\ &= 56 \text{ cm}^2 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$x = \frac{t^3}{3} - 2t^2 + 3t$$

$$\dot{x} = t^2 - 4t + 3$$

when $t = 0$

$$\dot{x} = 3 \text{ m/s}$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve $v = 0$, or equivalent merit	1

Sample answer:

Need $\dot{x} = 0$

$$t^2 - 4t + 3 = 0$$

$$(t - 3)(t - 1) = 0$$

$$t = 3, \quad t = 1$$

Question 12 (d) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds the time when $a = 0$, or equivalent merit	1

Sample answer:

$$\ddot{x} = 2t - 4$$

$$2t - 4 = 0$$

$$2t = 4$$

$$t = 2$$

when $t = 2$

$$x = \frac{2^3}{3} - 2(2)^2 + 3(2)$$

$$= \frac{2}{3} \text{ m}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	3
• Finds the stationary points, or equivalent merit	2
• Attempts to solve $\frac{dy}{dx} = 0$, or equivalent merit	1

Sample answer:

$$y = 6x^2 - x^3$$

$$y' = 12x - 3x^2$$

$$y'' = 12 - 6x$$

stationary points occur when $y' = 0$

$$12x - 3x^2 = 0$$

$$3x(4 - x) = 0$$

$$3x = 0 \quad 4 - x = 0$$

$$x = 0 \quad x = 4$$

stationary points at (0, 0) and (4, 32)

when $x = 0$, $y'' = 12 > 0 \therefore$ minimum at (0,0)

$x = 4$, $y'' = -12 < 0 \therefore$ maximum at (4,32)

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows second derivative vanishes at $x = 2$, or equivalent merit	1

Sample answer:

Point of inflexion when $y'' = 0$

$$12 - 6x = 0$$

$$6x = 12$$

$$x = 2$$

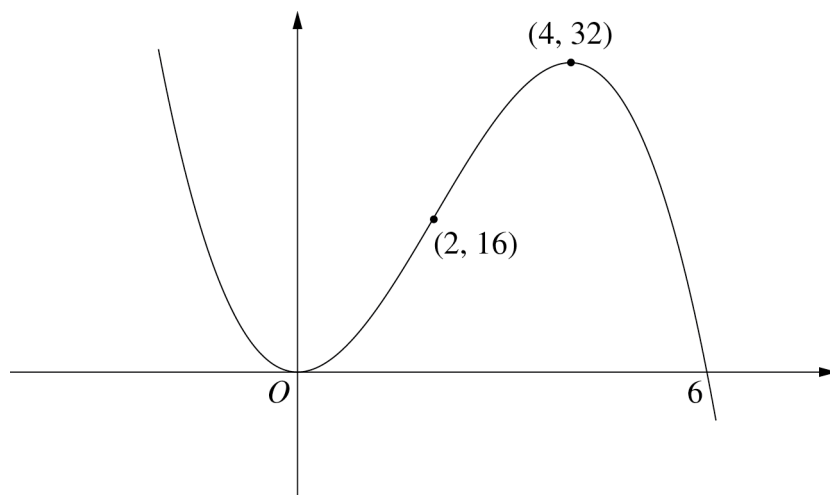
when $x = 2 - \epsilon$, $y'' > 0$

when $x = 2 + \epsilon$, $y'' < 0 \therefore (2, 16)$ is a point of inflexion

Question 13 (a) (iii)

Criteria	Marks
• Provides correct sketch	2
• Provides curve with correct shape, or equivalent merit	1

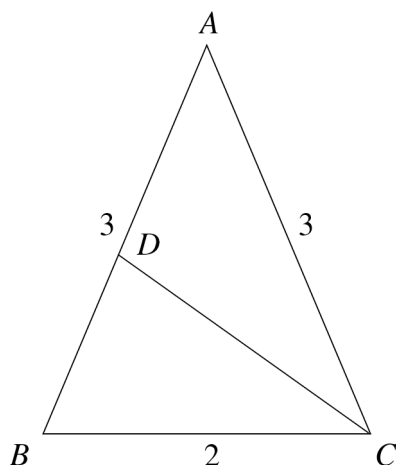
Sample answer:



Question 13 (b) (i)

Criteria	Marks
• Provides correct proof	2
• Finds one pair of congruent angles, or equivalent merit	1

Sample answer:



In $\triangle ABC$ and $\triangle CBD$ angle B is common.

Both triangles are isosceles so $\angle CDB = \angle CBD = \angle ABC = \angle ACB$ hence $\triangle ABC$ is similar to $\triangle CBD$ (AA).

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use ratios to find a correct equation involving BD , or equivalent merit	1

Sample answer:

$$AD = AB - BD$$

From part (i)

$$\frac{BD}{BC} = \frac{BC}{BA}$$

$$\frac{BD}{2} = \frac{2}{3}$$

$$BD = \frac{4}{3}$$

$$AD = 3 - \frac{4}{3}$$

$$= \frac{5}{3}$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains $184 = 92e^{50k}$, or equivalent merit	1

Sample answer:

$$p(t) = 92e^{kt}$$

$$\text{when } t = 50, \quad p = 184$$

$$92e^{50k} = 184$$

$$e^{50k} = 2$$

$$50k = \ln(2)$$

$$k = \frac{1}{50} \ln(2)$$

$$= 0.01386294361$$

$$\doteq 0.0139$$

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses $t = 110$, or equivalent merit	1

Sample answer:

When $t = 110$, $p = ?$

$$p = 92e^{110 \times 0.0139}$$

$$= 424.4476077$$

$$\div 424$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$A = \frac{1}{2} \times 3 \times 6 \times \sin 60^\circ$$

$$= \frac{9\sqrt{3}}{2} \text{ square units}$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds the area of another triangle, or equivalent merit	1

Sample answer:

$$\frac{1}{2} \times 3 \times x \times \sin 30^\circ + \frac{1}{2} \times 6 \times x \times \sin 30^\circ = \frac{9\sqrt{3}}{2}$$

$$\frac{3x}{2} + 3x = 9\sqrt{3}$$

$$\frac{9x}{2} = 9\sqrt{3}$$

$$x = 2\sqrt{3}$$

Question 14 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains correct integral for volume in terms of y only, or equivalent merit	2
• Attempts to use an integral of the form $\pi \int x^2 dy$, or equivalent merit	1

Sample answer:

$$y = x^4 + 1$$

$$x^4 = y - 1$$

$$x^2 = \sqrt{y-1}$$

$$V = \pi \int_1^{10} x^2 dy$$

$$= \pi \int_1^{10} (y-1)^{\frac{1}{2}} dy$$

$$= \pi \left[\frac{2(y-1)^{\frac{3}{2}}}{3 \times 1} \right]_1^{10}$$

$$= \frac{2\pi}{3} \left(9^{\frac{3}{2}} - 0 \right)$$

$$= \frac{2\pi}{3} \times 27$$

Volume = 18π cubic units

Question 14 (c)

Criteria	Marks
• Provides correct solution	3
• Obtains correct discriminant, or equivalent merit	2
• Obtains correct derivative, or equivalent merit	1

Sample answer:

$$f(x) = x^3 + kx^2 + 3x - 5$$

$$f'(x) = 3x^2 + 2kx + 3$$

For stationary points need $f'(x) = 0$

$$\Delta = (2k)^2 - 4 \times 3 \times 3$$

$$= 4k^2 - 36$$

For no stationary points

$$\Delta < 0$$

$$4k^2 - 36 < 0$$

$$k^2 - 9 < 0$$

$$(k + 3)(k - 3) < 0$$

$$\therefore -3 < k < 3$$

Question 14 (d) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$T_1 = 2^1 + 1$$

$$= 3$$

$$T_2 = 2^2 + 2$$

$$= 6$$

$$T_3 = 2^3 + 3$$

$$= 11$$

Question 14 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly sums one series, or equivalent merit	1

Sample answer:

$$(2^1 + 1) + (2^2 + 2) + (2^3 + 3) + \dots + (2^n + n)$$

$$= (2^1 + 2^2 + 2^3 + \dots + 2^n) + (1 + 2 + 3 + \dots + n)$$

For $2 + 2^2 + 2^3 + \dots + 2^n$

$$a = 2, r = 2, n = 20$$

$$S_{20} = \frac{2(2^{20} - 1)}{2 - 1}$$

$$= 2097150$$

For $1 + 2 + 3 + \dots + n$

$$a = 1, d = 1, n = 20$$

$$S_{20} = \frac{20}{2}(1 + 20)$$

$$= 210$$

$$\therefore \text{Total} = 2097150 + 210$$

$$= 2097360$$

Question 14 (e) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\text{Probability} = 1 - 0.9 \times 0.95$$

$$= 1 - 0.855$$

$$= 0.145$$

Question 14 (e) (ii)

Criteria	Marks
• Provides correct solution	2
• Demonstrates knowledge of complementary events, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \text{Probability} &= 0.5 \times 0.9 \times 0.9 + 0.5 \times 0.95 \times 0.95 \\
 &= 0.405 + 0.45125 \\
 &= 0.85625
 \end{aligned}$$

Question 15 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Let $t = 0$

$$\begin{aligned}
 L(0) &= 12 + 2\cos\left(\frac{2\pi \times 0}{366}\right) \\
 &= 12 + 2\cos 0 \\
 &= 12 + 2 \\
 &= 14
 \end{aligned}$$

Day is 14 hours long

Question 15 (a) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 -1 &\leq \cos\left(\frac{2\pi t}{366}\right) \leq 1 \\
 -2 &\leq 2\cos\left(\frac{2\pi t}{366}\right) \leq 2
 \end{aligned}$$

Shortest length of daylight when $2\cos\left(\frac{2\pi t}{366}\right) = -2$

$$L(t) = 12 - 2 = 10 \text{ hours}$$

Shortest length of daylight is 10 hours.

Question 15 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains $\cos\left(\frac{2\pi t}{366}\right) = -\frac{1}{2}$, or equivalent merit	1

Sample answer:

$$11 = 12 + 2\cos\left(\frac{2\pi t}{366}\right)$$

$$-1 = 2\cos\left(\frac{2\pi t}{366}\right)$$

$$-\frac{1}{2} = \cos\left(\frac{2\pi t}{366}\right)$$

$$\frac{2\pi t}{366} = \frac{2\pi}{3} \quad \text{or} \quad \frac{4\pi}{3}$$

$$\frac{2\pi t}{366} = \frac{2\pi}{3}$$

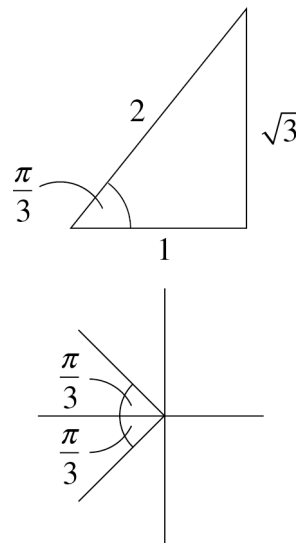
$$t = \frac{366}{3}$$

$t = 122$ days after 21 December

$$\frac{2\pi t}{366} = \frac{4\pi}{3}$$

$$t = \frac{2 \times 366}{3}$$

$= 244$ days after 21 December



Question 15 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains correct equation involving logarithms, or equivalent merit	2
• Obtains correct equation involving integrals, or equivalent merit	1

Sample answer:

$$\int_0^k \frac{dx}{x+3} + \int_k^{45} \frac{dx}{x+3} = \int_0^{45} \frac{dx}{x+3}$$

$$\int_0^k \frac{dx}{x+3} = \int_k^{45} \frac{dx}{x+3}$$

$$2 \int_0^k \frac{dx}{x+3} = \int_0^{45} \frac{dx}{x+3}$$

$$2 \left[\ln(x+3) \right]_0^k = \left[\ln(x+3) \right]_0^{45}$$

$$2 \left[\ln(k+3) - \ln 3 \right] = \ln(48) - \ln 3$$

$$\frac{(k+3)^2}{9} = 16$$

$$(k+3)^2 = 144$$

$$k+3 = \pm 12 \quad (k > 0)$$

$$k+3 = 12$$

$$k = 9$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains correct integral for the area, or equivalent merit	1

Sample answer:

$$A = \int_0^3 2x - (x^3 - 7x) dx$$

$$A = \int_0^3 2x - x^3 + 7x dx$$

$$= \int_0^3 9x - x^3 dx$$

$$= \left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_0^3$$

$$= \left(\frac{9 \times 9}{2} - \frac{81}{4} \right) - (0 - 0)$$

$$= \frac{81}{2} - \frac{81}{4}$$

$$= \frac{81}{4}$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses correct function values in Simpson's Rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \int_0^3 9x - x^3 dx &= \frac{3-0}{6} \left[f(0) + 4f\left(\frac{3}{2}\right) + f(3) \right] \\
 &= \frac{1}{2} \left[0 + 4 \left[\frac{9 \times 3}{2} - \left(\frac{3}{2}\right)^3 \right] + 0 \right] \\
 &= \frac{1}{2} \times 4 \times \left(\frac{27}{2} - \frac{27}{8} \right) \\
 &= 2 \times \left(\frac{4 \times 27}{8} - \frac{27}{8} \right) \\
 &= 2 \times \frac{81}{8} \\
 &= \frac{81}{4}
 \end{aligned}$$

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains the equation $3x^2 - 7 = 2$, or equivalent merit	1

Sample answer:

P is on curve $y = x^3 - 7x$ where tangent to curve is parallel to $y = 2x$.

$$\frac{dy}{dx} = 3x^2 - 7 \quad \text{slope of line} = 2$$

want slope of tangent to be 2

$$3x^2 - 7 = 2$$

$$3x^2 = 9$$

$$x^2 = 3$$

$$x = \pm\sqrt{3} \quad \text{want } x > 0 \text{ so}$$

$$x = +\sqrt{3}$$

$$\begin{aligned} \text{when } x = +\sqrt{3} \quad y &= x^3 - 7x = (\sqrt{3})^3 - 7 \times \sqrt{3} \\ &= 3\sqrt{3} - 7\sqrt{3} \\ &= -4\sqrt{3} \end{aligned}$$

Coordinates of P are $(\sqrt{3}, -4\sqrt{3})$

Question 15 (c) (iv)

Criteria	Marks
• Provides correct solution	2
• Finds the distance from p to the line, or equivalent merit	1

Sample answer:

Perpendicular distance from P to $y = 2x$

$$\begin{aligned}
 d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\
 &= \frac{|2\sqrt{3} - 1 \times (-4\sqrt{3})|}{\sqrt{2^2 + (-1)^2}} \\
 &= \frac{|2\sqrt{3} + 4\sqrt{3}|}{\sqrt{5}} \\
 &= \frac{6\sqrt{3}}{\sqrt{5}}
 \end{aligned}$$

distance from $(0, 0)$ to $(3, 6)$

$$\begin{aligned}
 d &= \sqrt{(3-0)^2 + (6-0)^2} \\
 &= \sqrt{9+36} \\
 &= \sqrt{45} \\
 &= 3\sqrt{5}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of triangle} &= \frac{1}{2} \times \frac{6\sqrt{3}}{\sqrt{5}} \times 3\sqrt{5} \\
 &= 9\sqrt{3}
 \end{aligned}$$

Alternative solution

Take vertical distance between P and the line as base of two triangles.

$$\text{When } x = \sqrt{3}, \quad y = 2x \Rightarrow y = 2\sqrt{3}$$

$$\text{Vertical distance is } 2\sqrt{3} - (-4\sqrt{3}) = 6\sqrt{3}$$

$$A_1 = \frac{1}{2}b \times h_1 = \frac{1}{2} \times 6\sqrt{3} \times \sqrt{3}$$

$$A_2 = \frac{1}{2}b \times h_2 = \frac{1}{2} \times 6\sqrt{3} \times (3 - \sqrt{3})$$

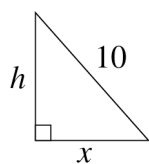
$$\begin{aligned} A &= \frac{1}{2} \times 6\sqrt{3} \times [\sqrt{3} + (3 - \sqrt{3})] \\ &= 3\sqrt{3} \times 3 = 9\sqrt{3} \end{aligned}$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$V = \frac{1}{3}\pi r^2 h$$



$$10^2 = h^2 + x^2$$

$$h = \sqrt{100 - x^2}$$

; radius = x

$$V = \frac{1}{3}\pi x^2 \sqrt{100 - x^2}$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly applies product rule, or equivalent merit	1

Sample answer:

$$V = \frac{1}{3}\pi x^2 \sqrt{100 - x^2}$$

$$u = \frac{1}{3}\pi x^2 \quad V = (100 - x^2)^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{2}{3}\pi x \quad \frac{dV}{dx} = \frac{1}{2} \times -2x \times (100 - x^2)^{-\frac{1}{2}}$$

$$= \frac{-x}{\sqrt{100 - x^2}}$$

$$\frac{dV}{dx} = \frac{-\pi x^3}{3\sqrt{100 - x^2}} + \frac{2\pi x \sqrt{100 - x^2}}{3}$$

$$= \frac{-\pi x^3 + 2\pi x(100 - x^2)}{3\sqrt{100 - x^2}}$$

$$= \frac{-\pi x^3 + 200\pi x - 2\pi x^3}{3\sqrt{100 - x^2}}$$

$$= \frac{200\pi x - 3\pi x^3}{3\sqrt{100 - x^2}}$$

$$= \frac{\pi x(200 - 3x^2)}{3\sqrt{100 - x^2}}$$

Question 16 (a) (iii)

Criteria	Marks
• Provides correct solution	3
• Verifies that $x = \frac{10\sqrt{6}}{3}$ gives the maximum, or equivalent merit	2
• Finds the values of x at the stationary points, or equivalent merit	1

Sample answer:

$$\frac{dV}{dx} = 0$$

$$\frac{\pi x(200 - 3x^2)}{3\sqrt{100 - x^2}} = 0$$

$$\pi x(200 - 3x^2) = 0$$

$$\pi x = 0, \quad 200 - 3x^2 = 0$$

$$x = 0 \quad 3x^2 = 200$$

Not a solution, $x = \pm \sqrt{\frac{200}{3}}$

$$x > 0 \quad x > 0$$

$$x = \frac{10\sqrt{2}}{\sqrt{3}}$$

$$x = \frac{10\sqrt{6}}{3}$$

Nature ?

x	8	$\frac{10\sqrt{6}}{3}$	9
$\frac{dV}{dx}$	11.1701	0	-92.974

$\therefore$ Maximum volume when $x = \frac{10\sqrt{6}}{3}$

$$\ell = r\theta$$

$$\ell = 10\theta$$

$$\therefore 2\pi r = 10\theta$$

$$2\pi x = 10\theta$$

$$\theta = \frac{\pi x}{5}$$

Therefore

$$\theta = \frac{\pi}{5} \times \frac{10\sqrt{6}}{3}$$

$$\theta = \frac{2\sqrt{6}\pi}{3}$$

Question 16 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Enumerates some of the possibilities, or equivalent merit	1

Sample answer:

Cannot win if there are no numbers between the first two.

Numbers between

	1	2	3	4	5	6
1	×	×	1	2	3	4
2	×	×	×	1	2	3
3	1	×	×	×	1	2
4	2	1	×	×	×	1
5	3	2	1	×	×	×
6	4	3	2	1	×	×

So there are 16 rolls that cannot win before third roll.

$$\text{Probability} = \frac{16}{36} = \frac{4}{9}$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly deals with one case, or equivalent merit	1

Sample answer:

If 1 number between

$$\text{probability of win is } \frac{1}{36} \times \frac{1}{6}$$

8 cases

2 number between

$$\text{probability is } \frac{1}{36} \times \frac{2}{6}$$

6 cases

3 number between

$$\text{probability is } \frac{1}{36} \times \frac{3}{6}$$

4 cases

4 number between

$$\text{probability is } \frac{1}{36} \times \frac{4}{6}$$

2 cases

$$P(\text{win}) = 8 \times \frac{1}{36} \times \frac{1}{6} + 6 \times \frac{1}{36} \times \frac{2}{6} + 4 \times \frac{1}{36} \times \frac{3}{6} + 2 \times \frac{1}{36} \times \frac{4}{6}$$

$$= \frac{1}{216} (8 + 12 + 12 + 8)$$

$$= \frac{40}{216}$$

$$= \frac{5}{27}$$

Question 16 (c) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$A_1 = 300\,000(1.04) - P$$

$$A_2 = A_1(1.04) - P(1.05)$$

$$= 300\,000(1.04)^2 - P(1.04) - P(1.05)$$

$$= 300\,000(1.04)^2 - P[(1.04) + (1.05)]$$

Question 16 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$A_3 = A_2(1.04) - P(1.05)(1.05)$$

$$= 300\,000(1.04)^3 - P(1.04)^2 - P(1.04)(1.05) - P(1.05)^2$$

$$= 300\,000(1.04)^3 - P[(1.04)^2 + (1.04)(1.05) + (1.05)^2]$$

Question 16 (c) (iii)

Criteria	Marks
• Provides correct solution	3
• Sums the series for A_n , or equivalent merit	2
• Obtains an expression for A_n , or equivalent merit	1

Sample answer:

Continuing this pattern

$$A_n = 300\,000(1.04)^n - P \left[(1.04)^{n-1} + (1.04)^{n-2}(1.05)^1 + (1.04)^{n-3}(1.05)^2 + \dots + (1.05)^{n-1} \right]$$

The second term is a geometric series with n terms, $a = (1.04)^{n-1}$ and $r = \frac{1.05}{1.04} = \frac{105}{104}$

$$\begin{aligned} &= 300\,000(1.04)^n - P(1.04)^{n-1} \left[\frac{\left(\frac{105}{104}\right)^n - 1}{\left(\frac{105}{104}\right) - 1} \right] \\ &= 300\,000(1.04)^n - P(1.04)^{n-1} \frac{\left[\left(\frac{105}{104}\right)^n - 1\right]}{\left(\frac{1}{104}\right)} \\ &= 300\,000(1.04)^n - 104P(1.04)^{n-1} \left[\left(\frac{105}{104}\right)^n - 1 \right] \end{aligned}$$

For money to be in the account we require $A_n > 0$

$$\text{ie } 300\,000(1.04)^n > 100P(1.04)^n \left[\left(\frac{105}{104}\right)^n - 1 \right]$$

$$\therefore \left(\frac{105}{104}\right)^n - 1 < \frac{300\,000}{100P}$$

$$\therefore \left(\frac{105}{104}\right)^n < 1 + \frac{3000}{P}$$

2018 HSC Mathematics Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.1	P3
2	1	6.7	P4
3	1	6.1	P4
4	1	6.5	P4
5	1	12.5, 13.5	H5
6	1	3.3	H5
7	1	11.4	H8
8	1	9.3	P4
9	1	10.8	H6, H7
10	1	13.3, 13.6	H5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	1.1	P3
11 (b)	2	1.2	P3
11 (c)	2	1.3	P3
11 (d) (i)	1	7.1	H5
11 (d) (ii)	2	7.1	H5
11 (e)	2	12.5	H3, H5
11 (f)	2	8.8, 13.5	H5
11 (g)	2	8.8, 12.4	H3, H5
12 (a) (i)	1	5.4	P4
12 (a) (ii)	2	5.5	P4
12 (b)	3	6.2, 13.5	H5
12 (c) (i)	2	2.3, 2.4	P4
12 (c) (ii)	2	2.3, 2.5	H5
12 (d) (i)	1	14.3	H5
12 (d) (ii)	2	14.3	H5
12 (d) (iii)	2	14.3	H5
13 (a) (i)	3	10.2	H6
13 (a) (ii)	2	10.2	H6
13 (a) (iii)	2	10.5	H6
13 (b) (i)	2	2.3	P4
13 (b) (ii)	2	2.3	P4
13 (c) (i)	2	14.2	H3, H4
13 (c) (ii)	2	14.2	H3, H4

Question	Marks	Content	Syllabus outcomes
14 (a) (i)	1	5.3, 5.5	P3
14 (a) (ii)	2	5.3, 5.5	P3
14 (b)	3	11.4	H8
14 (c)	3	8.9, 9.1, 9.2, 10.2	P4, P5, P7, H6
14 (d) (i)	1	7.5	H5
14 (d) (ii)	2	7.5	H5
14 (e) (i)	1	3.3	H5
14 (e) (ii)	2	3.3	H5
15 (a) (i)	1	13.2	H4
15 (a) (ii)	1	13.2	H4
15 (a) (iii)	2	5.3, 13.2	H4
15 (b)	3	11.4, 12.2, 12.5	H8
15 (c) (i)	2	11.4	H8
15 (c) (ii)	2	11.3	H8
15 (c) (iii)	2	10.7	H6
15 (c) (iv)	2	6.5	P4
16 (a) (i)	1	2.3	P3
16 (a) (ii)	2	8.8	P7
16 (a) (iii)	3	10.6, 13.1	H5
16 (b) (i)	2	3.1, 3.2, 3.3	H5
16 (b) (ii)	2	3.1, 3.2, 3.3	H5
16 (c) (i)	1	7.5	H5
16 (c) (ii)	1	7.5	H5
16 (c) (iii)	3	7.5	H5